

Parametric Equations & Modeling

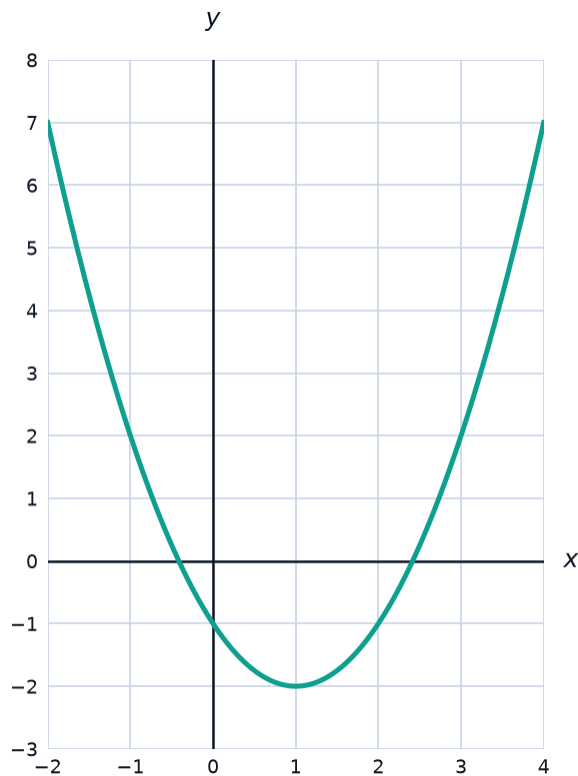


Evaluating parametric equations

- 1 Given $x(t) = t^2 - 1$ and $y(t) = 2t + 3$, find the point on the curve at $t = 2$.
 - 2 Given $x(t) = 3\cos t$ and $y(t) = 3\sin t$, find the point at $t = \frac{\pi}{3}$, and identify the shape traced by the curve.
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Eliminating the parameter

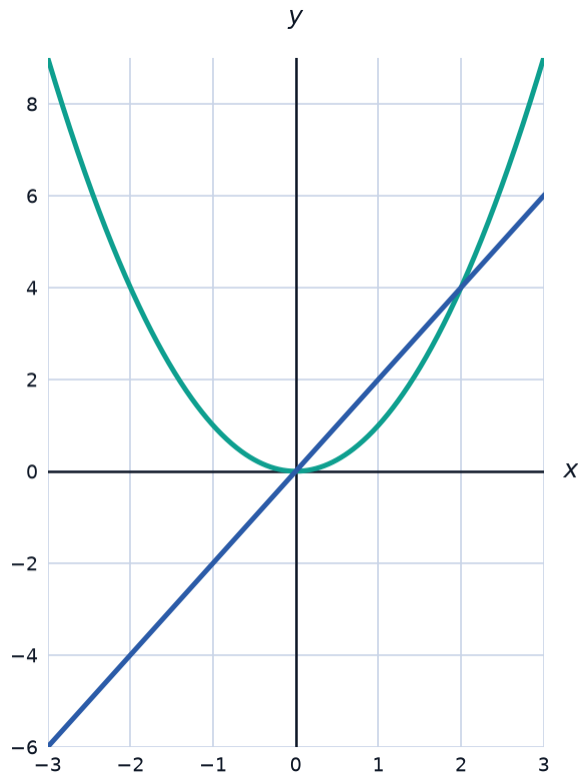
- 3 Eliminate the parameter from $x(t) = t + 1$, $y(t) = t^2 - 2$ to find the rectangular equation, and identify the curve.



- 4 Eliminate the parameter from $x(t) = 2t - 3$, $y(t) = 4t + 1$ to find the rectangular equation.
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Visualizing a parametric curve's components

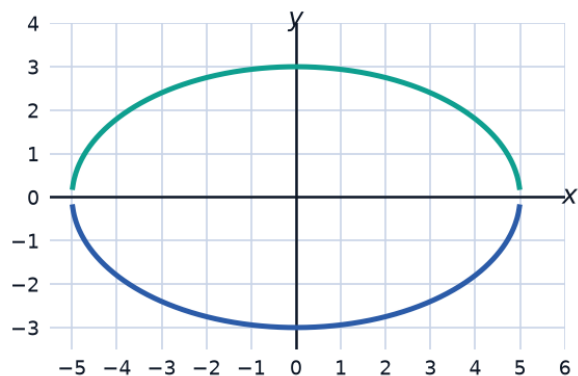
- 5 The graph shows the two component functions $x(t) = t^2$ (upper curve) and $y(t) = 2t$ (lower, straight line) for the parametric curve $x(t) = t^2$, $y(t) = 2t$, plotted separately against t on the horizontal axis.



- a Eliminate the parameter t to find the rectangular equation relating x and y for this curve.
- b Use your rectangular equation to describe the shape of the actual parametric curve (not the component graphs shown).

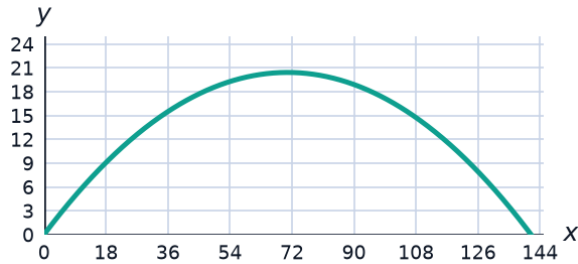
Parametric equations for lines and conics

- 6 Write parametric equations for the line segment from $(1, 2)$ to $(5, 10)$, using $t \in [0, 1]$.
- 7 Write parametric equations for an ellipse with semi-major axis 5 (horizontal) and semi-minor axis 3 (vertical), centered at the origin.



Modeling projectile motion parametrically

- 8 A ball is launched with initial speed 40 m/s at an angle of 30° above horizontal. Its position is modeled by $x(t) = 40\cos(30^\circ)t$ and $y(t) = 40\sin(30^\circ)t - 4.9t^2$. Find the ball's position at $t = 2$ seconds, to one decimal place.



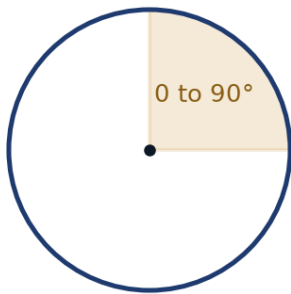
- 9 For the same projectile, find the time at which the ball lands (returns to $y = 0$).

Comparing parametric speed and direction

- 10 Two objects follow the same path $y = x^2$, but object A uses $x(t) = t$, $y(t) = t^2$ while object B uses $x(t) = 2t$, $y(t) = 4t^2$. Explain why they trace the same curve but do not move through it at the same rate.

Restricting the parameter's domain

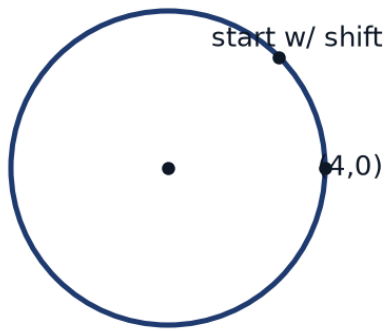
- 11 The parametric equations $x(t) = \cos t$, $y(t) = \sin t$ trace the full unit circle for $t \in [0, 2\pi)$. Describe the portion of the circle traced when the domain is restricted to $t \in \left[0, \frac{\pi}{2}\right]$.



- 12 Explain why restricting t in $x(t) = t^2$, $y(t) = t$ to $t \geq 0$ changes the rectangular relation from $x = y^2$ (a full sideways parabola) to only its upper half.

Parametric equations for circular motion with a phase shift

- 13 A point moves along a circle of radius 4 according to $x(t) = 4\cos\left(t + \frac{\pi}{4}\right)$, $y(t) = 4\sin\left(t + \frac{\pi}{4}\right)$. Find its position at $t = 0$, and explain the effect of the $\frac{\pi}{4}$ phase shift compared to $x(t) = 4\cos t$, $y(t) = 4\sin t$.



Multiple choice — easy 15

- 14 Given $x(t) = -3t - 2$ and $y(t) = -5t$, find the point on the curve at $t = -2$.
- a. (10, 4)
 - b. (4, 10)
 - c. (4, 11)
 - d. (5, 10)
- 15 Given $x(t) = -2t - 5$ and $y(t) = 2t + 6$, find the point on the curve at $t = -5$.
- a. (6, -4)
 - b. (-4, 5)
 - c. (5, -3)
 - d. (5, -4)

- 16 Given $x(t) = -5t - 5$ and $y(t) = 2t - 3$, find the point on the curve at $t = 2$.
- a. $(-15, 2)$
 - b. $(-14, 1)$
 - c. $(-15, 1)$
 - d. $(1, -15)$
- 17 Given $x(t) = -6t - 3$ and $y(t) = -4t - 5$, find the point on the curve at $t = 3$.
- a. $(-20, -17)$
 - b. $(-21, -17)$
 - c. $(-21, -16)$
 - d. $(-17, -21)$
- 18 Given $x(t) = -2t + 9$ and $y(t) = 3t + 5$, find the point on the curve at $t = 0$.
- a. $(9, 5)$
 - b. $(9, 6)$
 - c. $(10, 5)$
 - d. $(5, 9)$
- 19 Given $x(t) = 2t - 5$ and $y(t) = -2t + 2$, find the point on the curve at $t = 1$.
- a. $(-3, 0)$
 - b. $(-3, 1)$
 - c. $(-2, 0)$
 - d. $(0, -3)$

- 20 Given $x(t) = -3t + 4$ and $y(t) = -3t + 4$, find the point on the curve at $t = -3$.
- a. (13, 13)
 - b. (11, 15)
 - c. (14, 13)
 - d. (13, 14)
- 21 Given $x(t) = -t + 5$ and $y(t) = -5t - 7$, find the point on the curve at $t = 0$.
- a. $(-7, 5)$
 - b. $(5, -6)$
 - c. $(5, -7)$
 - d. $(6, -7)$
- 22 Given $x(t) = 4t - 7$ and $y(t) = t - 7$, find the point on the curve at $t = -5$.
- a. $(-27, -12)$
 - b. $(-26, -12)$
 - c. $(-27, -11)$
 - d. $(-12, -27)$
- 23 Given $x(t) = -t - 3$ and $y(t) = 5t - 6$, find the point on the curve at $t = 0$.
- a. $(-2, -6)$
 - b. $(-3, -6)$
 - c. $(-6, -3)$
 - d. $(-3, -5)$

- 24 Given $x(t) = 6t + 9$ and $y(t) = 2t + 5$, find the point on the curve at $t = -2$.
- a. $(-2, 1)$
 - b. $(-3, 1)$
 - c. $(-3, 2)$
 - d. $(1, -3)$
- 25 Given $x(t) = 2t + 4$ and $y(t) = -6t - 7$, find the point on the curve at $t = 3$.
- a. $(-25, 10)$
 - b. $(10, -25)$
 - c. $(11, -25)$
 - d. $(10, -24)$
- 26 Given $x(t) = 4t + 4$ and $y(t) = t + 9$, find the point on the curve at $t = -6$.
- a. $(-20, 4)$
 - b. $(-19, 3)$
 - c. $(3, -20)$
 - d. $(-20, 3)$
- 27 Given $x(t) = 3t - 5$ and $y(t) = t + 1$, find the point on the curve at $t = -6$.
- a. $(-22, -5)$
 - b. $(-23, -5)$
 - c. $(-23, -4)$
 - d. $(-5, -23)$

28 Given $x(t) = 6t - 4$ and $y(t) = 5t - 3$, find the point on the curve at $t = 5$.

a. $(27, 22)$

b. $(22, 26)$

c. $(26, 22)$

d. $(26, 23)$

Multiple choice — medium 15

29 Eliminate the parameter from $x(t) = t - 2$ and $y(t) = -3t + 4$ to find the rectangular equation for y in terms of x .

a. $y = -3x - 1$

b. $y = -3x + 4$

c. $y = 3x - 2$

d. $y = -3x - 2$

30 Eliminate the parameter from $x(t) = t - 9$ and $y(t) = -7t + 7$ to find the rectangular equation for y in terms of x .

a. $y = 7x - 56$

b. $y = -7x - 55$

c. $y = -7x + 7$

d. $y = -7x - 56$

- 31 Eliminate the parameter from $x(t) = t - 8$ and $y(t) = 7t - 9$ to find the rectangular equation for y in terms of x .
- a. $y = 7x - 9$
 - b. $y = -7x + 47$
 - c. $y = 7x + 47$
 - d. $y = 7x + 48$
- 32 Eliminate the parameter from $x(t) = t + 5$ and $y(t) = 7t - 2$ to find the rectangular equation for y in terms of x .
- a. $y = 7x - 36$
 - b. $y = 7x - 37$
 - c. $y = 7x - 2$
 - d. $y = -7x - 37$
- 33 Eliminate the parameter from $x(t) = t - 6$ and $y(t) = -5t - 4$ to find the rectangular equation for y in terms of x .
- a. $y = 5x - 34$
 - b. $y = -5x - 34$
 - c. $y = -5x - 33$
 - d. $y = -5x - 4$
- 34 Eliminate the parameter from $x(t) = t - 1$ and $y(t) = 8t - 5$ to find the rectangular equation for y in terms of x .
- a. $y = 8x - 5$
 - b. $y = -8x + 3$
 - c. $y = 8x + 3$
 - d. $y = 8x + 4$

- 35 Eliminate the parameter from $x(t) = t + 2$ and $y(t) = 6t - 1$ to find the rectangular equation for y in terms of x .
- a. $y = 6x - 1$
 - b. $y = -6x - 13$
 - c. $y = 6x - 12$
 - d. $y = 6x - 13$
- 36 Eliminate the parameter from $x(t) = t - 5$ and $y(t) = t - 7$ to find the rectangular equation for y in terms of x .
- a. $y = -x - 2$
 - b. $y = x - 7$
 - c. $y = x - 2$
 - d. $y = x - 1$
- 37 Eliminate the parameter from $x(t) = t + 3$ and $y(t) = -3t - 3$ to find the rectangular equation for y in terms of x .
- a. $y = 3x + 6$
 - b. $y = -3x + 6$
 - c. $y = -3x - 3$
 - d. $y = -3x + 7$
- 38 Eliminate the parameter from $x(t) = t + 6$ and $y(t) = t + 4$ to find the rectangular equation for y in terms of x .
- a. $y = x - 2$
 - b. $y = -x - 2$
 - c. $y = x - 1$
 - d. $y = x + 4$

- 39 Eliminate the parameter from $x(t) = t - 6$ and $y(t) = -3t - 3$ to find the rectangular equation for y in terms of x .
- a. $y = -3x - 20$
 - b. $y = 3x - 21$
 - c. $y = -3x - 21$
 - d. $y = -3x - 3$
- 40 Eliminate the parameter from $x(t) = t + 9$ and $y(t) = 3t + 4$ to find the rectangular equation for y in terms of x .
- a. $y = 3x - 23$
 - b. $y = 3x - 22$
 - c. $y = 3x + 4$
 - d. $y = -3x - 23$
- 41 Eliminate the parameter from $x(t) = t$ and $y(t) = 8t - 9$ to find the rectangular equation for y in terms of x .
- a. $y = 8x - 9$
 - b. $y = -8x - 9$
 - c. $y = 8x - 7$
 - d. $y = 8x - 8$
- 42 Eliminate the parameter from $x(t) = t - 2$ and $y(t) = -8t + 3$ to find the rectangular equation for y in terms of x .
- a. $y = -8x + 3$
 - b. $y = -8x - 12$
 - c. $y = 8x - 13$
 - d. $y = -8x - 13$

- 43 Eliminate the parameter from $x(t) = t - 2$ and $y(t) = -t - 5$ to find the rectangular equation for y in terms of x .
- a. $y = -x - 7$
 - b. $y = -x - 6$
 - c. $y = x - 7$
 - d. $y = -x - 5$
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Multiple choice — hard 15

- 44 Given $x(t) = t^2 + 6$ and $y(t) = -2t + 8$, find the point on the curve at $t = 0$.
- a. (8, 8)
 - b. (6, 8)
 - c. (8, 6)
 - d. (6, 10)
- 45 Given $x(t) = t^2 + 2$ and $y(t) = -2t + 7$, find the point on the curve at $t = -1$.
- a. (3, 9)
 - b. (9, 3)
 - c. (5, 9)
 - d. (3, 11)
- 46 Given $x(t) = t^2 + 4$ and $y(t) = 2t - 2$, find the point on the curve at $t = 4$.
- a. (20, 8)
 - b. (22, 6)
 - c. (6, 20)
 - d. (20, 6)

- 47 Given $x(t) = t^2 - 9$ and $y(t) = 4t + 3$, find the point on the curve at $t = -1$.
- a. $(-8, 1)$
 - b. $(-1, -8)$
 - c. $(-6, -1)$
 - d. $(-8, -1)$
- 48 Given $x(t) = t^2 + 9$ and $y(t) = -t - 7$, find the point on the curve at $t = 2$.
- a. $(13, -9)$
 - b. $(-9, 13)$
 - c. $(13, -7)$
 - d. $(15, -9)$
- 49 A curve is given by $x(t) = 4\cos t$, $y(t) = 4\sin t$. Find the exact point on the curve at $t = \frac{11\pi}{6}$ (330°).
- a. $(2\sqrt{3}, 2)$
 - b. $(2\sqrt{3}, -2)$
 - c. $(-2, 2\sqrt{3})$
 - d. $(-2\sqrt{3}, -2)$
- 50 A curve is given by $x(t) = 8\cos t$, $y(t) = 8\sin t$. Find the exact point on the curve at $t = \frac{5\pi}{4}$ (225°).
- a. $(-4\sqrt{2}, -4\sqrt{2})$
 - b. $(4\sqrt{2}, -4\sqrt{2})$
 - c. $(0, -8)$
 - d. $(-4\sqrt{2}, 4\sqrt{2})$

- 51 A curve is given by $x(t) = 8\cos t$, $y(t) = 8\sin t$. Find the exact point on the curve at $t = \frac{3\pi}{2}$ (270°).
- a. $(0, -8)$
 - b. $(-8, 0)$
 - c. $(-4\sqrt{3}, -4)$
 - d. $(0, 8)$
- 52 A curve is given by $x(t) = 4\cos t$, $y(t) = 4\sin t$. Find the exact point on the curve at $t = \frac{4\pi}{3}$ (240°).
- a. $(-2, 2\sqrt{3})$
 - b. $(2, -2\sqrt{3})$
 - c. $(-2\sqrt{3}, -2)$
 - d. $(-2, -2\sqrt{3})$
- 53 A curve is given by $x(t) = 4\cos t$, $y(t) = 4\sin t$. Find the exact point on the curve at $t = \frac{7\pi}{4}$ (315°).
- a. $(2\sqrt{2}, 2\sqrt{2})$
 - b. $(-2\sqrt{2}, 2\sqrt{2})$
 - c. $(-2\sqrt{2}, -2\sqrt{2})$
 - d. $(2\sqrt{2}, -2\sqrt{2})$
- 54 A projectile is launched with initial speed 34 m/s at 30° above horizontal, so $y(t) = 34\sin(30^\circ)t - 4.9t^2$. Find $y(2)$, to one decimal place.
- a. 12.4
 - b. 19.4
 - c. 16.4
 - d. 14.4

- 55 A projectile is launched with initial speed 49 m/s at 60° above horizontal, so $y(t) = 49\sin(60^\circ)t - 4.9t^2$. Find $y(2)$, to one decimal place.
- a. 65.3
 - b. 70.3
 - c. 67.3
 - d. 63.3
- 56 A projectile is launched with initial speed 45 m/s at 30° above horizontal, so $y(t) = 45\sin(30^\circ)t - 4.9t^2$. Find $y(2)$, to one decimal place.
- a. 23.4
 - b. 25.4
 - c. 30.4
 - d. 27.4
- 57 A projectile is launched with initial speed 60 m/s at 30° above horizontal, so $y(t) = 60\sin(30^\circ)t - 4.9t^2$. Find $y(1)$, to one decimal place.
- a. 27.1
 - b. 25.1
 - c. 23.1
 - d. 30.1
- 58 A projectile is launched with initial speed 59 m/s at 60° above horizontal, so $y(t) = 59\sin(60^\circ)t - 4.9t^2$. Find $y(4)$, to one decimal place.
- a. 128.0
 - b. 126.0
 - c. 124.0
 - d. 131.0